



Systems Engineering

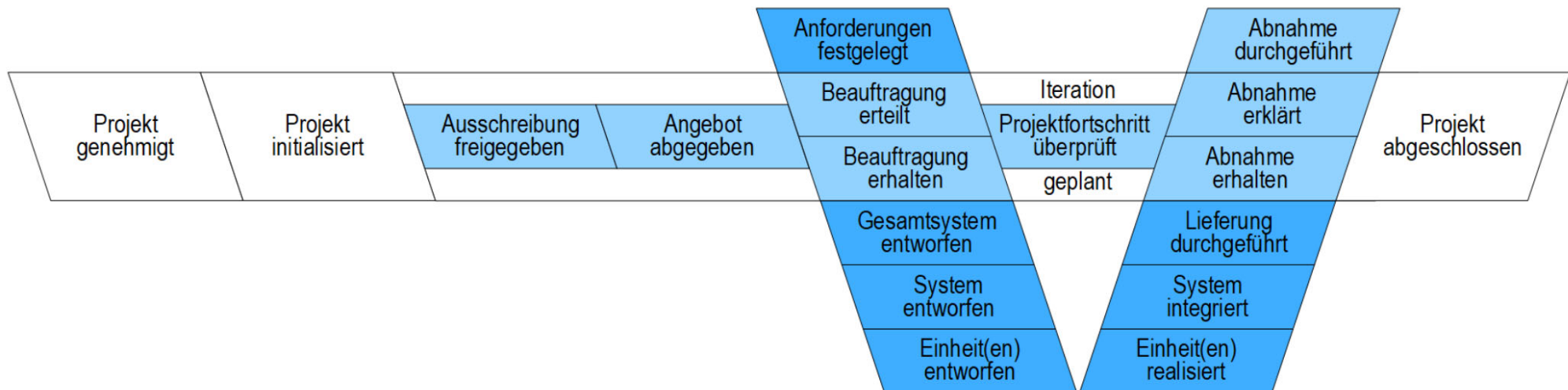
Technical Systems Engineering Process

Prof. Dr. Kai Höfig



V-Modell XT

- ◆ The V-Modell XT (For extreme tailoring) was formerly developed by the German armed forces and can be adopted to many different **IT projects and IT systems**.
- ◆ As a process model, it shows what activities should be performed, but not how. That means a development process can look different, e.g agile.
- ◆ This model is strongly order-based.

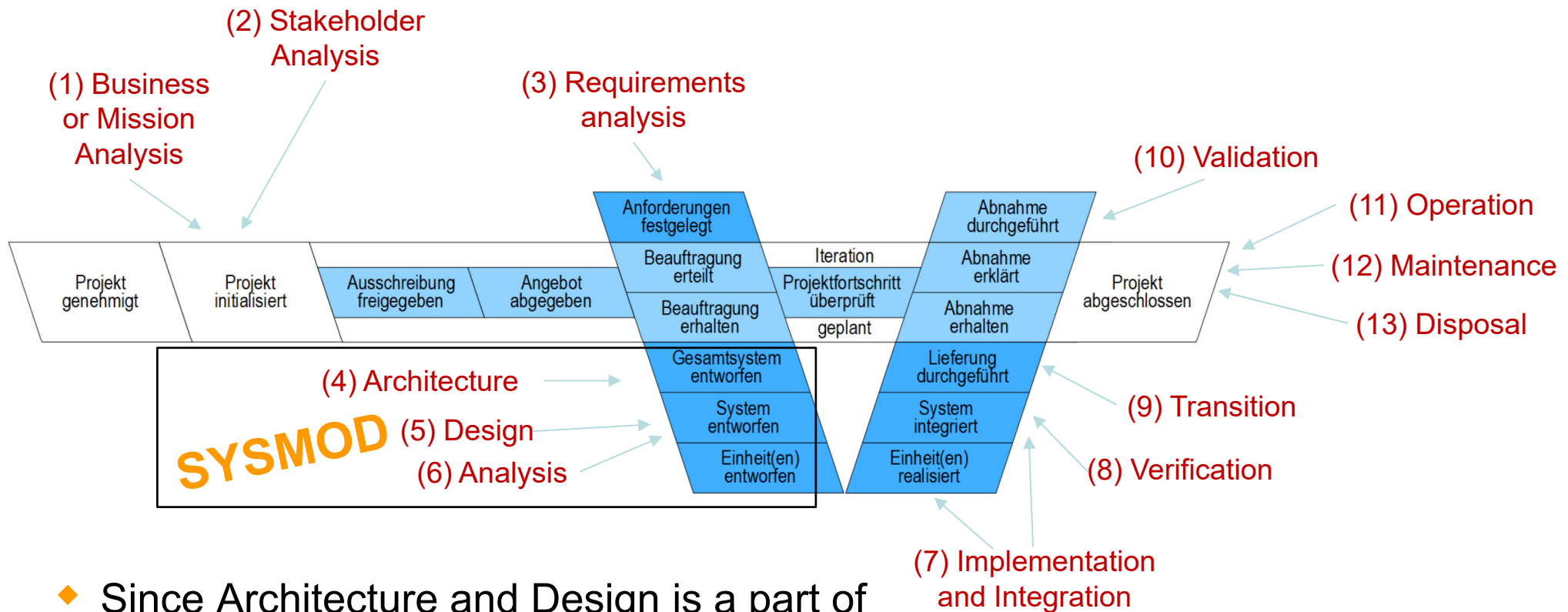


<http://ftp.tu-clausthal.de/pub/institute/informatik/v-modell-xt/Releases/2.3/V-Modell-XT-Gesamt.pdf>



Technical Systems Engineering Process(es)

- ◆ The technical systems engineering process as proposed in **INCOSE'S SE Handbook** in correlation with **SWE V-Modell XT**.



- ◆ Since Architecture and Design is a part of the systems engineering process with intense modelling activities, there is a special implementation for it in SysML: **SYSMOD**



Technical process – Preparation – Step 1 & 2

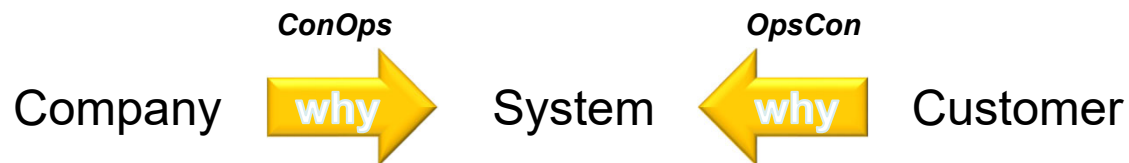
- ◆ **Business or mission analysis (process step 1)**

Strategy for analysis, define problem or opportunity, characterize solution space, evaluate alternatives, define requirements from a business perspective and define how these requirements are validated.

- ◆ **ConOps**, *Concept of operation: Intended way of business operation referring to one or more systems. What is our company and why do we have a system?*



OpsCon, *System operational concept: What a system does and why does it make sense from a user's perspective. In contrast to ConOps, why do the people using a system, need it.*

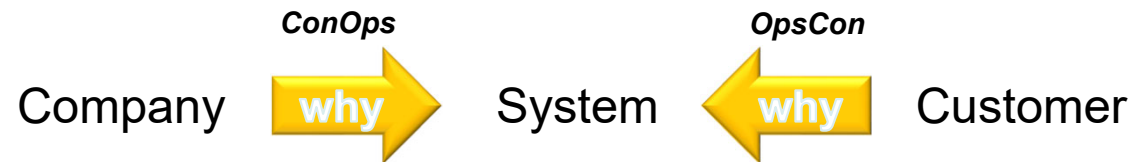


- ◆ **Stakeholder Needs (process step 2)**

Define stakeholder needs, derive requirements and organization, development of lifecycle concepts



The Candy Grabber



Stakeholder Needs (2)

Define stakeholder needs, derive requirements and organization, development of lifecycle concepts



System requirements definition process (process step 3)

- ◆ The central activity of this phase is, to transform business and stakeholder needs into a technical view that meets the operational view of the user

- ◆ Good requirements are always
 - **Necessary**
 - **Implementation independent**
 - Unambiguous
 - Complete
 - Singular
 - Achievable
 - Verifiable
 - Confirming
 - Complete
 - Consistent
 - Feasible and affordable
 - Bounded
- ◆ ..and enable/checking for
 - **Trace to parent**
 - Trace to source
 - Trace to interface definition
 - **Trace to peer requirements**
 - Trace to verification method
 - Trace to verification requirement(s)
 - Trace to verification results
 - Verification status
 - Validation status
 - Priority
 - Criticality
 - Risk
 - Key driving requirement
 - Owner
 - Rationale
 - Applicability
 - Type – e.g. input, output, interface, reliability (can also be tags)



Architecture definition process (process step 4)

- ◆ In this step, different **architecture alternatives** are compared to each other to find the most suitable that meets customer requirements and project constraints.
- ◆ The main activity here is to develop a **model** for each alternative that consists of different views.
- ◆ The **views** shall enable a measurement of the architecture and allow a validation with the customer.

Part of this lecture (SysML) using the SYSMOD
Architecture and Design Modelling Process





Design definition process (process step 5)

- ◆ After an architecture alternative is found, that seems to meet all requirements and constraints, this architecture is extended to a **design** by adding more and more information to the model to a point, where enough information exists to implement the system.
- ◆ Also **alternatives** should be evaluated here and can result in the evidence, that the wrong architecture has been selected.

Part of this lecture (SysML) using the SYSMOD
Architecture and Design Modelling Process





System analysis process (process step 6)

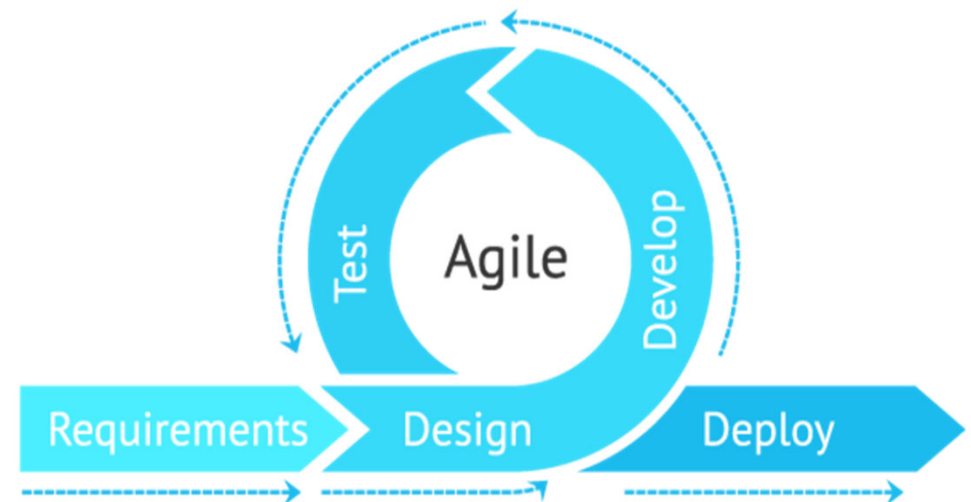
- ◆ The purpose of the system analysis process is to provide a rigorous basis of data and information for technical understanding to aid decision-making across the system's lifecycle containing
- ◆ **LCC** – Lifecycle costs including labor and non-labor costs, development, manufacturing, service realization, sales, customer utilization, supply chain, maintenance and disposal costs.
- ◆ **Risk analysis**, that does not focus on the development risks but on the technical risks of a system through out its lifetime. Typically, also probabilities and impact of risks are evaluated.
- ◆ **Effectiveness analysis** shows to which degree the system will meet its operational properties through out the lifetime.

*Part of this
lecture
(Reliability
Analysis)*



Implementation and integration processes (process step 7)

- ◆ The **implementation process** has the purpose to realize a specified system element.
- ◆ During the **integration process**, the system elements of the implementation process are synthesized into a realization of a system that satisfies system requirements, architecture and design.





Verification process (process step 8)

- ◆ The purpose of verification is to provide objective evidence that either a system or a system element fulfills the specified requirements and characteristics.
- ◆ The following verification techniques are basic techniques
 - Inspection – typically what can be observed by humans
 - Analysis – theoretical compliance without any intervention such as simulation
 - Demonstration – usecases that are observed by humans
 - Test – typically executed and evaluated by automations
 - Analogy or similarity – existing evidence is used from a different element
 - Simulation – evidence is based on models or mock-ups
 - Sampling – verification of characteristics using samples

**Part of this
lecture (Formal
verification) and
also part of SKS**



Test

- ◆ Testing is the comparison of expected results (also called the test oracle) with obtained results.
- ◆ Unit tests are test cases of a comparatively small focus of the system and performed on system element level or unit level.
- ◆ System integration tests are performed on system scope and are executed on the system synthesized of the system elements (that ideally passed unit tests)

***Part of SKS, SE
and other***



Transition process (process step 9)

- ◆ During the transition process, the custody of the system is transferred from the development team to the customer. One could also say delivery, but many systems are build on promise and are never really delivered.





Validation (process step 10)

- ◆ The purpose of the validation process is to provide objective evidence that the system, when in use, fulfills its business or mission objectives and stakeholder requirements, achieving its intended use in its intended operational environment.
- ◆ Includes assessment of OpsCon
- ◆ Follows a validation plan, that is created throughout the development
- ◆ Validation is applied to the operational system, but also during the development using simulations, prototypes and analysis.
- ◆ The stakeholders should play an important role during validation, that also include operators, end users, management, owners and so on.



Operation (process step 11)

- ◆ The purpose of this process is to **use the system** to deliver its service.
- ◆ It includes
 - Preparation for operation – make equipment, personnel and services available, perform safety training, establish work schedule, establish feedback loop
 - Perform operation – operate according to OpsCon, track performance
 - Manage results – document the results of operation record anomalies and analyze them
 - Customer support – establish processes to receive customer requests

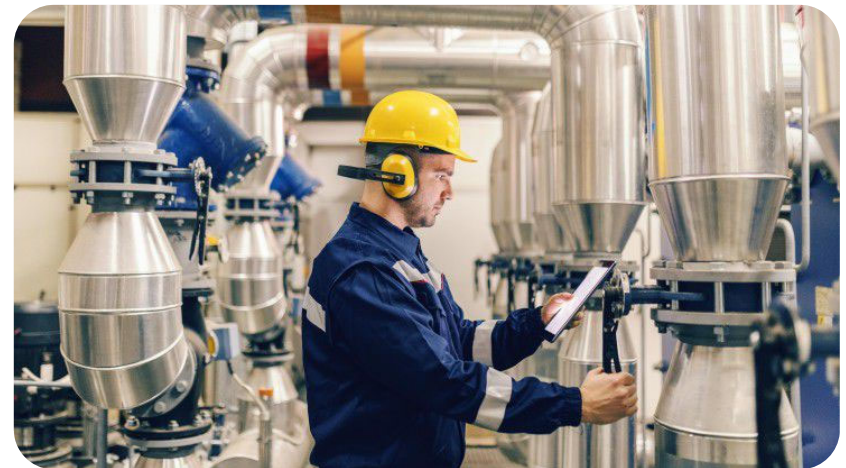




Maintenance (process step 12)

- ♦ The purpose of this process is to **sustain the capability** of the system to provide its service
- ♦ **Predictive maintenance** – anticipate maintenance activities before they are necessary
- ♦ **Preventive maintenance** – anticipate maintenance activities before they are necessary to prevent risks
- ♦ **Reliability-centered maintenance** – anticipate maintenance activities before they are necessary to prevent risks of unreliability

Part of this
lecture
(Reliability
Analysis)





Disposal (process step 13)

- ◆ The purpose of this process is to end the existence of a system or a system element for a specified intended use, appropriately handle replaced or retired elements and to properly attend to identified critical disposal needs.
- ◆ Disposal strategy and design considerations should be an integral part of system design and development.
- ◆ ISO 14000 series includes standards for lifecycle assessment

